

6	(a) unit of magnetic flux density field normal to (straight) conductor carrying current of 1 A force per unit length is 1 N m^{-1}	B1 M1 A1 [3]
(b)	(i) force on particle always normal to direction of motion (and speed of particle is constant) magnetic force provides the centripetal force	M1 A1 [2]
	(ii) $mv^2/r = Bqv$ $r = mv/Bq$	M1 A0 [1]
(c)	(i) the momentum/speed is becoming less so the radius is becoming smaller	M1 A1 [2]
	(ii) 1. spirals are in opposite directions so oppositely charged	M1 A1 [2]
	2. equal <u>initial</u> radii so equal (initial) speeds	M1 A1 [2]

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7 (a) (i) packet/quantum of energy

M1